

Elizabeth Dietrich

University of California, Berkeley

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Position

University of California, Berkeley

GRADUATE STUDENT RESEARCHER

Electrical Engineering and Computer Sciences

Berkeley, USA

08/2023 – Today

Interests

Methods

reachability analysis · formal methods · data-driven control

Topics

autonomous systems · control theory · statistical learning

Education

University of California, Berkeley

PHD IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCES

Advisor: Murat Arcak

Berkeley, CA, USA

08/2023 – Today

Indiana University, Bloomington

B.S. IN COMPUTER SCIENCE, MINOR IN MATHEMATICS

Thesis: A beginner's guide to Iris, Coq and separation logic, Advisor: Dennis Shasha

Bloomington, IN, USA

08/2017 – 05/2021

Visits

Norwegian University of Science and Technology (NTNU)

VISITING RESEARCHER

MC Lab (Asgeir J Sørensen and Roger Skjetne)

Trondheim, Norway

06/2025 – 05/2026

Norwegian University of Science and Technology (NTNU)

VISITING RESEARCHER

MC Lab (Asgeir J Sørensen)

Trondheim, Norway

05/2024 – 07/2024

Publications

* first author(s)

PREPRINTS

[P1] Varun Madabushi*, **Elizabeth Dietrich***, Hanna Krasowski, and Maegan Tucker. *Finite-Step Invariant Sets for Hybrid Systems with Probabilistic Guarantees*. arXiv:2604.05102. 2026

JOURNAL ARTICLES

[J1] Hanna Krasowski*, **Elizabeth Dietrich***, Emir Cem Gezer, Roger Skjetne, Asgeir Johan Sørensen, and Murat Arcak. “pac-STL: PAC-Bounded Signal Temporal Logic from Data-Driven Reachability Analysis”. In: *Accepted in IEEE Control Systems Letters* (2026)

CONFERENCE PAPERS

[C5] **Elizabeth Dietrich***, Hanna Krasowski*, and Murat Arcak. “Probably Approximately Correct (PAC) Guarantees for Data-Driven Reachability Analysis: A Theoretical and Empirical Comparison”. In: *Accepted for Proc. of the IEEE Conference on Decision and Control (CDC)*. 2026

[C4] **Elizabeth Dietrich***, Hanna Krasowski, Vegard Flovik, and Murat Arcak. “Importance Sampling for Statistical Certification of Viable Initial Sets”. In: *Accepted for Proc. of the IEEE Conference on Decision and Control (CDC)*. 2026

- [C3] **Elizabeth Dietrich***, Rosalyn Devonport, Stephen Tu, and Murat Arcak. “Data-Driven Reachability with Scenario Optimization and the Holdout Method”. In: *2025 IEEE 64th Conference on Decision and Control (CDC)*. 2025, pp. 3925–3931
- [C2] **Elizabeth Dietrich***, Emir Cem Gezer*, Bingzhuo Zhong, Murat Arcak, Majid Zamani, Roger Skjetne, and Asgeir Johan Sørensen. “Symbolic Control for Autonomous Docking of Marine Surface Vessels”. In: *2025 IEEE 64th Conference on Decision and Control (CDC)*. 2025, pp. 3021–3026
- [C1] **Elizabeth Dietrich***, Alex Devonport, and Murat Arcak. “Nonconvex scenario optimization for data-driven reachability”. In: *Proceedings of the 6th Annual Learning for Dynamics Control Conference*. Vol. 242. Proceedings of Machine Learning Research. PMLR, 15–17 Jul 2024, pp. 514–527

Invited Talks

- 2025** NSF Frontier Review, UC Berkeley · SFI Autoship Review, NTNU · Marine Cybernetics Seminar, NTNU · ANSR Seminar, UC Berkeley, online
- 2022** Informatics, Computing, and Engineering Seminar, IUB · Intelligent Systems Symposium, JHU APL

Mentoring

STUDENT PROJECTS

- 2025- **NSF IRES Project on Autonomous Marine Operations for the Arctic** UCB
Athul Krishnan & Chloe Yejin Lynn

STUDENT THESES

- 2025- **Master’s Thesis: Safe Reinforcement Learning for Intelligent Ship Guidance and Control** NTNU & UCB
Petter Dalhaug

Professional Experience

- 08/2021 – 07/2023 **Associate Staff: Research Scientist** · Johns Hopkins University Applied Physics Laboratory
05/2020 – 08/2021 **Undergraduate Research Assistant** · New York University, Courant Institute
05/2019 – 08/2019 **Software Engineer Intern** · Amazon

Teaching Experience

- Spring 2025 **Optimization of Engineering Systems (MAS-E 238C)**, Graduate Teaching Assistant (online)
Fall 2024 **Analysis and Control of Nonlinear Systems (MAS-E 250A)**, Graduate Teaching Assistant (online)
Spring 2022 **MATLAB II**, STEM Academy Instructor JHUAPL
Fall 2020 **Training for Paraprofessionals in Student Affairs (EDUC-U208)**, Undergraduate Teaching Assistant
Fall 2018 & 2019 **Introduction to Computer Science (CSCI-C211)**, Undergraduate Teaching Assistant

Service to the Profession

COMMITTEES & ORGANIZATIONS

- 2024 - **EECS Equal Access to Application Assistance (EEAAA)**, Assist underrepresented prospective PhD students on their graduate school applications to help equalize access to guidance on the application process. UCB
- 2026 **EECS PhD Visit Days Coordinator**, Coordinate visit days for close to 200 students admitted to the EECS PhD program. UCB
- 02/2026 **Application reviewer for Berkeley EECS MEng program**, Read and evaluate applications for admission to Berkeley’s EECS professional master programs. UCB
- 03/2025 **EECS PhD CIR & CPSDA Visit Days**, Organization of activities for prospective PhD students in controls, automation, learning, and embedded systems. UCB

PEER REVIEW

Conferences (reviewed 4 submissions): Conference on Decision and Control (CDC)

Journals (reviewed 4 submissions): Control System Letters (L-CSS), Nonlinear Analysis: Hybrid Systems (NAHS)

Awards, Fellowships, & Grants

2026	D-Lab Data Science & AI Fellowship , University of California, Berkeley
2026	EECS Departmental Chair's Graduate Award , University of California, Berkeley
2025	Fernström Fellowship , University of California, Berkeley
2023	NSF Graduate Research Fellowship (GRFP) , National Science Foundation
2021	Kate Hevner Mueller Senior Award , Indiana University, Bloomington
2020	Goldwater Scholar , Barry Goldwater Scholarship and Excellence in Education Foundation